

DEEPL - Lab 8

Attention Mechanism

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Introduction (brief, see lecture)

In this lab, we will explore attention mechanisms that allow neural networks to focus on a certain number of inputs : for example, to determine whether a sentiment is present in a sentence or not, it is not necessary to look at all the words but only at the parts that express sentiments.

Data and Model Preparation

You will use the IMDB dataset which contains 50,000 reviews from the Internet Movie Database (IMDb), and Glove word embeddings (see the code provided for the lab). The goal is to classify a review as either "positive" (pos) or "negative" (neg); the performance metric is the classification accuracy.

1 Baseline Model

Question 1

You will implement a simple model based on embeddings (figure 1) : a text $t = (t_1, \dots, t_n)$ is represented by the average of the embeddings $\mathbf{x}_i$ of the words it contains, i.e.

$$\hat{\mathbf{t}} = \frac{1}{n} \sum_i \mathbf{x}_i$$

Use a simple linear model with cross-entropy loss.

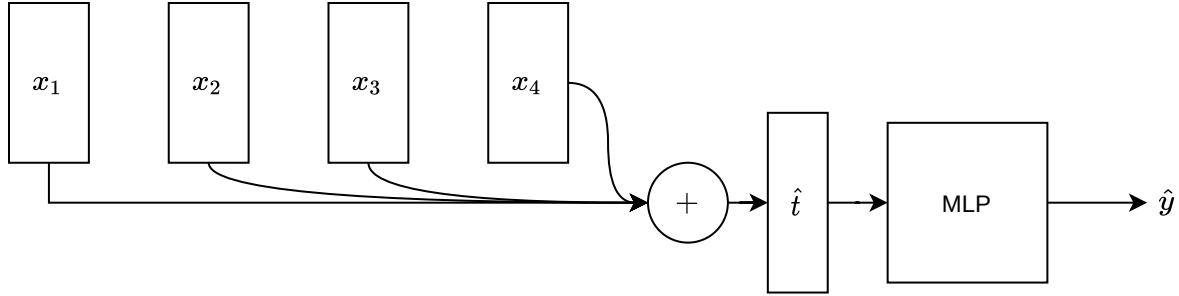


FIGURE 1 – Generic Pooling

2 Simple Attention

The previous model gives poor results because all words are used with the same weight to compute the barycentric representation of the text. The objective of this first very simple attention model (figure 2) is to strongly weight the words that are of interest for the classification task in order to build the weighted representation of the text :

$$\hat{\mathbf{t}} = \sum_i p(\mathbf{a}_i|\mathbf{t}) \mathbf{x}_i$$

with the coefficients $p(\mathbf{a}_i|\mathbf{t})$ representing the probability of attending to the i -th token of the text. We then use a linear classifier as before.

We propose to model these probabilities as follows :

$$\log p(\mathbf{a}_i|\mathbf{t}) = \text{constant} + \mathbf{q} \cdot \mathbf{x}_i$$

with $\mathbf{q}$ a vector called **query** : it allows expressing the relevance of the i -th token with the question posed. This vector will be learned during the training phase (it is an embedding of the query).

Question 2

Implement the model. Think about which simple function allows obtaining $p(\mathbf{a}_i|t)$ from $\mathbf{q} \cdot \mathbf{x}_i$.

During training, compute the histogram of the entropy of the attention weights. During inference, observe for specific examples which words receive attention, which allows seeing which word(s) were important for the decision.

Vocabulary point and global attention : word embeddings can be viewed as keys $\mathbf{k}_i$ from which we seek the most similar to the posed query $\mathbf{q}$. Each key is associated with a

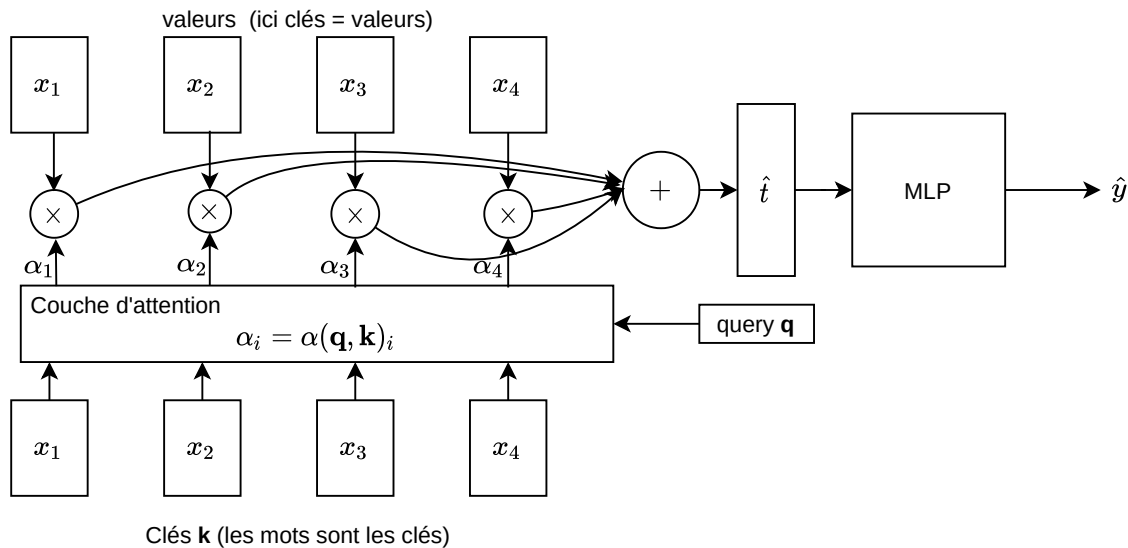


FIGURE 2 – Attention Model

value (in our case the key and value are the same - the word embedding) and this value is weighted by the similarity between the query and the key - the attention paid to the i -th value. We generally denote the attention vector $\alpha(\mathbf{q}, \mathbf{k})$.

3 Query and Value

We may want to make the query $\mathbf{q}$ depend on the text itself : for example, the vocabulary for talking about one's feelings about a horror film will not be the same as for romance films (though...).

Question 3

To do this, you will compute the query $\mathbf{q}$ as a function of the text by using the averaged text representation $\hat{\mathbf{t}}_m$ (exercice 1) and a linear transformation (you can make this model more complex as you wish depending on your results).

Then use this query to compute the document representation with

$$\log p(\mathbf{a}_i | \mathbf{t}) = \text{constante} + \mathbf{q}(\hat{\mathbf{t}}_m) \cdot \mathbf{x}_i$$

Similarly, it can be interesting to change the representation of a token for the decision by changing the *value* of a word embedding :

$$\hat{\mathbf{t}} = \sum_i p(\mathbf{a}_i|\mathbf{t})v_{\theta}(\mathbf{x}_i)$$

4 Optional : Contextual Models and Attention

Question 4

Create a new model that combines an LSTM or GRU to create *contextualized* embeddings. You can use one or more layers of LSTM/GRU. The attention layer this time focuses on the sequence of states from the last recurrent layer. As in the previous questions, we use a final linear layer for classification that takes as input the weighted sum of states.

5 Optional : Entropy

Question 5

In order to force attention to focus on a part of the input, it is possible to use an entropy criterion. Modify the attention models by adding this regularization and observe its effect.